Dynamic Guideline: WHO-Based Schistosomiasis - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Latest core WHO schistosomiasis guideline

- **Guideline title:** *WHO guideline on control and elimination of human schistosomiasis*
- **Publication date:** 14 February 2022
- **Publisher:** World Health Organization, Control of Neglected Tropical Diseases and Guidelines Review Committee
- **Scope:** Human schistosomiasis due to *Schistosoma mansoni, S. japonicum, S. mekongi, S. guineensis, S. intercalatum,* and *S. haematobium*
- **Target users:** Ministries of health, national neglected tropical disease programmes, schistosomiasis programme managers, clinicians, public-health officers, implementing partners, donors, and researchers
- **Target population:** People living in endemic communities; all at-risk age groups from 2 years old; infected individuals of any age; pregnant women after the first trimester; lactating women; preschool-aged children; school-aged children; adults at occupational or domestic freshwater exposure risk
- **Primary objective:** Support countries to control morbidity, eliminate schistosomiasis as a public-health problem, and move toward interruption of transmission under the WHO neglected tropical diseases road map 2021–2030 WHO guideline page, WHO guideline PDF.

Exactly 5 WHO schistosomiasis-related guideline/guidance documents synthesized

No.	WHO guideline / technical guidance document	Date	Role in this report	GRADE relevance
1	*WHO guideline on control and elimination of human schistosomiasis*	14 February 2022	Core guideline and latest WHO schistosomiasis guideline	Formal GRADE approach used NCBI Bookshelf
2	*Prevention and control of schistosomiasis and soil-*	2002 / WHA strategy	Historical foundation for preventive	Pre-GRADE WHO technical consensus;

No.	WHO guideline / technical guidance document	Date	Role in this report	GRADE relevance
	transmitted helminthiasis, WHO Technical Report Series No. 912	context from 2001	chemotherapy and combined morbidity-control strategy	ratings below are adapted WHO TRS 912
3	*Assessing the efficacy of anthelminthic drugs against schistosomiasis and soil-transmitted helminthiases*	18 June 2013	Drug-efficacy assessment and monitoring methods	Technical-methods guidance; adapted GRADE WHO technical document
4	*Assessing schistosomiasis and soil-transmitted helminthiases control programmes: monitoring and evaluation framework*	25 October 2024; WHO release 31 October 2024	Post-guideline programme monitoring and evaluation framework	Programme guidance; adapted GRADE WHO news, Google Books
5	WHO Technical Report Series No. 1064, *Selection and use of essential medicines*, including arpraziquantel recommendation for children	2025	Essential-medicine policy update for preschool-aged children	Evidence-based expert recommendation; adapted GRADE WHO TRS No. 1064

Requested WHO guideline-development case study clarification

The most recent WHO item matching the requested title format is **"Meaningful youth engagement in a WHO guideline development process: case study of WHO abortion care guideline (2022)"**, listed by WHO on **11 December 2025** and launched through a WHO webinar the same day WHO webinar page, WHO abortion topic page. It is **not schistosomiasis-specific** and is therefore not counted among the five schistosomiasis WHO guidance documents above.

1.2 Key Recommendations

WHO 2022 core recommendations for diagnosis, treatment, control, and elimination

Recommendation	Operational detail	GRADE Evidence Quality	Strength
Annual preventive chemotherapy in	Provide a single dose of praziquantel annually at	**Moderate** overall; preventive	**Strong**

Recommendation	Operational detail	GRADE Evidence Quality	Strength
endemic communities with ≥10% Schistosoma prevalence	**≥75% treatment coverage** to **all age groups from 2 years old**, including adults, pregnant women after the first trimester, and lactating women	chemotherapy reduces infection prevalence with moderate-to-high certainty; morbidity evidence by age group lower certainty	
Use a 10% prevalence threshold to initiate community preventive chemotherapy	Communities with *Schistosoma* prevalence ≥10% should receive preventive chemotherapy; below 10%, programmes should use more targeted approaches depending on prior programme history	**Very Low** for the exact threshold	**Conditional**
In low-prevalence areas, use test-and-treat or reduced-frequency continuation depending on programme history	Areas below 10% without prior regular preventive chemotherapy should use clinical test-and-treat; areas already receiving regular preventive chemotherapy may continue at same or reduced frequency toward interruption	**Very Low**	**Conditional**
Use annual praziquantel as the default frequency; consider escalation where response is inadequate	Annual treatment is standard; twice-yearly treatment may be needed in persistent hotspots or settings with limited response despite adequate coverage	**Moderate** for annual frequency; lower certainty for escalation decisions	**Strong for annual; Conditional for escalation**

Recommendation	Operational detail	GRADE Evidence Quality	Strength
Ensure health-facility access to praziquantel for all infected individuals	Treat all infected people regardless of age, including lactating women and pregnant women except in the first trimester; for children under 2 years, base treatment on testing and clinical judgement	**Moderate**	**Strong**
Combine chemotherapy with WASH, environmental management, behaviour change, and snail control	Preventive chemotherapy alone is insufficient for elimination; programmes should integrate safe water, sanitation, hygiene education, environmental modification, and focal snail control	**Low** for WASH and snail-control effect estimates, but strong biological and programmatic rationale	**Conditional**
Strengthen diagnostic surveillance for humans, snails, animal reservoirs, and low-transmission settings	Use microscopy, antigen testing, molecular tools, and surveillance approaches appropriate to transmission intensity and programme phase	**Low to Very Low** for many elimination-setting diagnostic decisions	**Conditional**

Sources: WHO guideline PDF, WHO publication page, ECRI guideline profile PDF, ESPEN practical assessment manual.

Practical diagnosis and clinical treatment synthesis

Clinical / programme question	Recommendation	GRADE Evidence Quality	Strength
Suspected intestinal schistosomiasis	Examine stool for eggs, commonly by Kato-Katz or equivalent microscopy; consider repeated sampling where feasible	Moderate for specificity; Low for sensitivity in light infection	Strong
Suspected urogenital schistosomiasis	Examine urine for eggs; filtration is the standard method for *S. haematobium*; urine	Moderate for egg detection specificity; Low for sensitivity in	Strong

Clinical / programme question	Recommendation	GRADE Evidence Quality	Strength
	reagent strips for haematuria can support field screening in children	low-intensity infection	
Travelers or migrants with light infection risk	Use serology when egg shedding is inconsistent; interpret antibodies as exposure evidence because serology does not distinguish past from active infection	Low	Conditional
Treatment of confirmed infection	Use praziquantel as first-line therapy for all major human *Schistosoma* species	High to Moderate	Strong
Timing in travelers	Treat at least **6–8 weeks after last freshwater exposure** so adult worms are present and immune response is mature	Moderate	Strong
Incomplete response or light infection	Consider repeat praziquantel after **2–4 weeks**	Low	Conditional
Species-based dosing in individual care	Common dosing: **40 mg/kg/day in 2 divided doses for 1 day** for *S. mansoni, S. haematobium,* and *S. intercalatum*; **60 mg/kg/day in 3 divided doses for 1 day** for *S. japonicum* and *S. mekongi*	Moderate	Strong

Sources: CDC Yellow Book, CDC Clinical Care, CDC Clinical Overview, WHO fact sheet, Medscape.

1.3 Guideline Development Methodology

- **Evidence review process:** The WHO 2022 guideline used PICO questions, commissioned and identified systematic reviews, and synthesized evidence on preventive chemotherapy, praziquantel safety, prevalence thresholds, treatment frequency, snail control, WASH, and diagnostics for humans, snails, animal reservoirs, and environmental assessment WHO guideline PDF.
- **GRADE methodology:** WHO applied GRADE to rate certainty as **High, Moderate, Low, or Very Low** and used an evidence-to-decision framework considering benefits, harms, resources, equity, acceptability, feasibility, values, and preferences NCBI Bookshelf.
- **Consensus method:** A WHO Guideline Development Group reviewed evidence and formulated recommendations. A later review reported that the WHO guideline produced **six recommendations** supported by **nine systematic reviews** Infectious Diseases of Poverty / Springer Nature.

- **Conflict of interest management:** WHO guideline processes included declarations of interests and management procedures; the 2024 monitoring and evaluation framework also included an annex on declarations of interest Google Books.
- **Values and implementation priorities:** WHO placed high value on prevention of morbidity, elimination as a public-health problem, feasibility across health-system contexts, and equity for groups previously under-reached by school-based treatment, including preschool-aged children and adults NCBI Bookshelf.

II. Evidence Summary After WHO Latest Guideline Release (WHO guideline on control and elimination of human schistosomiasis) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search Strategy

Evidence after the 14 February 2022 WHO guideline was assessed from WHO documents, authoritative clinical sources, peer-reviewed literature, systematic reviews, diagnostic studies, programme reports, and clinical-trial registries through **29 April 2026**.

Databases and sources represented in the provided evidence summaries

- WHO publications, WHO IRIS, WHO news, WHO fact sheets, WHO Essential Medicines documents
- WHO AFRO / ESPEN programme documents
- CDC Yellow Book and CDC clinical guidance
- ClinicalTrials.gov
- Peer-reviewed journals and indexed summaries including *Lancet Infectious Diseases*, *Lancet Microbe*, *Scientific Reports*, *Tropical Medicine and Infectious Disease*, *Frontiers*, and PMC-hosted articles
- Pediatric Praziquantel Consortium, EDCTP / Global Health EDCTP3, and Global Schistosomiasis Alliance resources

Inclusion criteria

- Published or listed after 14 February 2022
- Relevant to schistosomiasis diagnosis, praziquantel or arpraziquantel treatment, preventive chemotherapy, surveillance, monitoring and evaluation, WASH, snail control, elimination, or programme management
- Human schistosomiasis or directly relevant transmission-control evidence

Exclusion criteria

- Non-human-only studies without clear programme relevance
- Opinion pieces without actionable clinical or public-health implications

- Sources unrelated to schistosomiasis, except the requested WHO youth-engagement case-study clarification

2.2 Key New Studies

Study / source	Design	Key findings	GRADE Quality	Recommendation Strength / Impact
WHO 2024 monitoring and evaluation framework	WHO programme guidance, endorsed by WHO Technical Advisory Group in March 2024	Provides structured monitoring and evaluation for schistosomiasis and STH programmes, including programme phases, data-driven decisions, moderate/heavy infection intensity indicators, and optional morbidity indicators such as haematuria, blood in stool, and ultrasound lesions	Low for direct outcome effect; Moderate for programme relevance	Conditional: integrate into national programme monitoring
WHO 2025 Essential Medicines update: arpraziquantel	WHO expert committee technical report	Recommended adding arpraziquantel dispersible/orodispersible tablet to the WHO Model List of Essential Medicines for Children for preschool-aged children with schistosomiasis, based on similar efficacy and safety to praziquantel and improved paediatric administration	Moderate	Strong for inclusion in paediatric access planning
N'Goran et al. arpraziquantel Phase III trial,	Open-label, partly randomized Phase III trial	Arpraziquantel clinical programme treated 442 children; 352 with *S. mansoni* and 90 with *S.*	Moderate	Conditional-to-Strong: use where approved and

Study / source	Design	Key findings	GRADE Quality	Recommendation Strength / Impact
cited in WHO EML proposal	in children aged 3 months to 6 years in Côte d'Ivoire and Kenya	*haematobium*; supported efficacy, safety, and palatability in preschool-aged children		available for eligible children
EMA / WHO prequalification / Uganda introduction of arpraziquantel	Regulatory and implementation evidence	EMA positive scientific opinion in December 2023; WHO prequalification in May 2024; first preschool-aged child treated in Uganda in March 2025	Low for implementation outcomes; Moderate for regulatory confidence	Conditional: scale with pharmacovigilance and programme monitoring
ClinicalTrials.gov NCT06698510	Small-scale public-health intervention study	Evaluates introduction of arpraziquantel treatment for schistosomiasis control in preschool-aged children in endemic areas	Very Low until results available	Conditional: monitor results before broad implementation conclusions
Madagascar diagnostic accuracy study	Diagnostic accuracy study	Microscopy remains based on egg detection but has low sensitivity in low-intensity infection; evaluated POC-CCA, UCP-LF CAA, and real-time PCR as indicators of active infection	Low	Conditional: use multi-test strategies in low-intensity settings
2024 Scientific Reports study on Kato-Katz and CCA for S. japonicum	Diagnostic evaluation	Complementary tests improve diagnostic accuracy in low-transmission communities	Low	Conditional: combine diagnostics where elimination decisions require high confidence
2026 Pemba, Tanzania near-elimination diagnostic study	Prospective diagnostic study;	Evaluated six single-sample tests for *S. haematobium*, including PCR, RPA, UCP-LF	Very Low	Conditional: hypothesis-generating only

Study / source	Design	Key findings	GRADE Quality	Recommendation Strength / Impact
	medRxiv preprint	CAA, and urine filtration; UCP-LF CAA specificity reported as 96.4%, but evidence is not peer reviewed		
2024 review on chemical snail-vector control	Narrative / technical review	Reinforces integrated strategy: praziquantel alone affects only the human-to-snail pathway; snail control, WASH, environmental management, and health education target transmission more comprehensively	Low	Conditional: prioritize in hotspots and elimination settings
2024 Frontiers trial in female genital schistosomiasis	Phase 2 randomized controlled trial	Repeated versus single praziquantel dosing for female genital schistosomiasis showed no difference in efficacy in the available summary	Low	Conditional: do not routinely intensify dosing for FGS without clearer evidence
WHO 2024 treatment coverage data	Global programme surveillance	In 2024, 39.6% of people requiring schistosomiasis treatment globally were reached; 61.7% of school-aged children requiring preventive chemotherapy were treated	Low for causal effect; High relevance for programme gap	Strong programme priority: improve coverage toward ≥75%

Sources: WHO news, WHO TRS No. 1064, WHO EML proposal, Pediatric Praziquantel Consortium, ClinicalTrials.gov NCT06698510, Madagascar diagnostic accuracy study, Scientific Reports, medRxiv preprint, Tropical Medicine and Infectious Disease review, Frontiers, WHO fact sheet.

2.3 Evidence Gaps and Future Research Needs

Evidence gap	Why it matters	Current GRADE certainty	Research priority	Recommendation Strength
Optimal diagnostic algorithm in low-transmission and post-MDA settings	Microscopy misses low-intensity infection; antigen and molecular tests have variable performance	Low to Very Low	Comparative diagnostic accuracy studies using latent-class methods and field-feasible algorithms	Strong research priority
Verification of interruption of transmission	Elimination requires confidence that human, snail, environmental, and animal reservoirs are no longer sustaining transmission	Very Low	Standardized surveillance packages for humans, snails, environment, and animal reservoirs	Strong research priority
Best threshold for starting, reducing, or stopping preventive chemotherapy	The 10% threshold is operationally useful but supported by very low-certainty evidence	Very Low	Longitudinal programme studies comparing thresholds and stopping rules	Strong research priority
Arpraziquantel effectiveness at scale	Regulatory and Phase III data are promising, but real-world delivery, adherence, pharmacovigilance, and impact need monitoring	Low to Moderate	Implementation studies and post-introduction safety surveillance	Strong research priority
Female genital schistosomiasis treatment optimization	FGS causes substantial reproductive morbidity; best dosing, adjunctive anti-inflammatory therapy,	Low	Larger RCTs including clinical, gynecologic, and patient-centred outcomes	Strong research priority

Evidence gap	Why it matters	Current GRADE certainty	Research priority	Recommendation Strength
	and follow-up remain uncertain			
Integrated WASH and snail-control effectiveness	WHO recommends integration, but effect estimates are low certainty and context dependent	Low	Cluster trials, quasi-experimental studies, and implementation studies in hotspots	Strong research priority
Praziquantel supply and adult coverage	WHO reports limited praziquantel availability, especially for adults, and 2024 global coverage below the ≥75% target	Low for causal estimates; high programme relevance	Supply-chain, delivery-model, and community-engagement studies	Strong programme priority

2.4 Updated Recommendations Based on New Evidence

Original WHO 2022 Recommendation	New Evidence Impact After 14 February 2022	Updated Recommendation as of 29 April 2026	GRADE Quality	Strength
Annual praziquantel preventive chemotherapy at ≥75% coverage in communities with ≥10% prevalence	WHO 2024 coverage data show global treatment coverage remains below the WHO operational target; 39.6% of all people needing treatment and 61.7% of school-aged children	**Maintain WHO 2022 recommendation.** Intensify delivery models to reach ≥75% coverage across all eligible age groups from 2 years old, not only school-aged children	Moderate	Strong

Original WHO 2022 Recommendation	New Evidence Impact After 14 February 2022	Updated Recommendation as of 29 April 2026	GRADE Quality	Strength
	were reached in 2024			
Include all age groups from 2 years old in endemic communities	Arpraziquantel now provides a more age-appropriate paediatric option for younger children; WHO EMLc addition supports access planning	**Maintain and strengthen implementation.** Use standard praziquantel for eligible groups and introduce arpraziquantel for preschool-aged children where nationally approved, procured, and monitored	Moderate	Strong
Provide praziquantel access for all infected individuals regardless of age, with clinical judgement for children under 2 years	Arpraziquantel evidence supports treatment of children 3 months to 6 years where approved; safety below 3 months or <5 kg remains unestablished in WHO EML proposal	**Update operationally.** For children 3 months to 6 years, consider arpraziquantel according to approved label and national policy; for children under 3 months or <5 kg, use specialist judgement and avoid routine use absent evidence	Moderate for 3 months–6 years; Very Low below 3 months	Strong for approved ages; Conditional outside label
Use annual preventive chemotherapy, with escalation in poor-response settings	Persistent hotspots remain a concern; snail-control review supports combining chemotherapy with snail control and WASH rather than relying only	**Maintain annual default.** In persistent hotspots, first confirm coverage, adherence, diagnosis, and reinfection drivers; consider biannual treatment plus focal snail control, WASH, and behaviour change	Moderate for annual; Low for hotspot package	Strong for annual; Conditional for escalation

Original WHO 2022 Recommendation	New Evidence Impact After 14 February 2022	Updated Recommendation as of 29 April 2026	GRADE Quality	Strength
	on more frequent treatment			
Integrate WASH, environmental management, behaviour change, and snail control	2024 review reinforces biological rationale and importance of targeting both human-to-snail and snail-to-human transmission pathways	**Strengthen integration.** Prioritize snail control and WASH in persistent hotspots, high-transmission water-contact sites, and elimination settings	Low	Conditional
Use diagnostics to support mapping, monitoring, and elimination	New diagnostic studies support multi-test strategies; microscopy alone is insufficient in low-intensity settings	**Update diagnostic approach.** Use microscopy for routine confirmation where appropriate, but add antigen testing and/or molecular methods for low-transmission, traveler, migrant, and elimination-verification contexts	Low	Conditional
Monitor programme progress	WHO released a dedicated 2024 M&E framework aligned with 2030 targets	**Adopt the 2024 WHO M&E framework.** Integrate schistosomiasis data into national health information systems and use quality data for adaptive programme decisions	Low to Moderate	Strong programme recommendation

Actionable Clinical and Public-Health Algorithm

1. **Assess risk**
    - Determine residence, travel, freshwater exposure, occupation, pregnancy status, age, prior treatment, and local prevalence category.
2. **Diagnose**
    - Use stool microscopy for intestinal schistosomiasis and urine microscopy/filtration for urogenital schistosomiasis.
    - In travelers, migrants, low-intensity infection, or elimination settings, add serology, antigen testing, or molecular tools where available.
3. **Treat individual infection**
    - Use praziquantel as first-line therapy.
    - For travelers, treat at least 6–8 weeks after last exposure.
    - Consider repeat treatment after 2–4 weeks if infection is light, early, or response is uncertain.
4. **Treat endemic communities**
    - If community prevalence is ≥10%, provide annual single-dose praziquantel preventive chemotherapy to all age groups from 2 years old at ≥75% coverage.
5. **Include young children**
    - Use national policy and product availability to introduce arpraziquantel for eligible preschool-aged children where approved.
6. **Prevent reinfection and transmission**
    - Combine preventive chemotherapy with safe water, sanitation, hygiene education, environmental management, and focal snail control.
7. **Monitor and adapt**
    - Apply the WHO 2024 M&E framework, track coverage and infection intensity, identify persistent hotspots, and refine subdistrict targeting.

Overall conclusion: As of **29 April 2026**, the WHO 2022 guideline remains the core global guideline for schistosomiasis diagnosis, treatment, control, and elimination. Post-2022 evidence does **not** overturn the main WHO recommendations; instead, it strengthens implementation priorities: broader treatment coverage, paediatric arpraziquantel introduction where approved, stronger M&E, integrated snail/WASH interventions, and improved diagnostics for low-transmission and elimination settings.